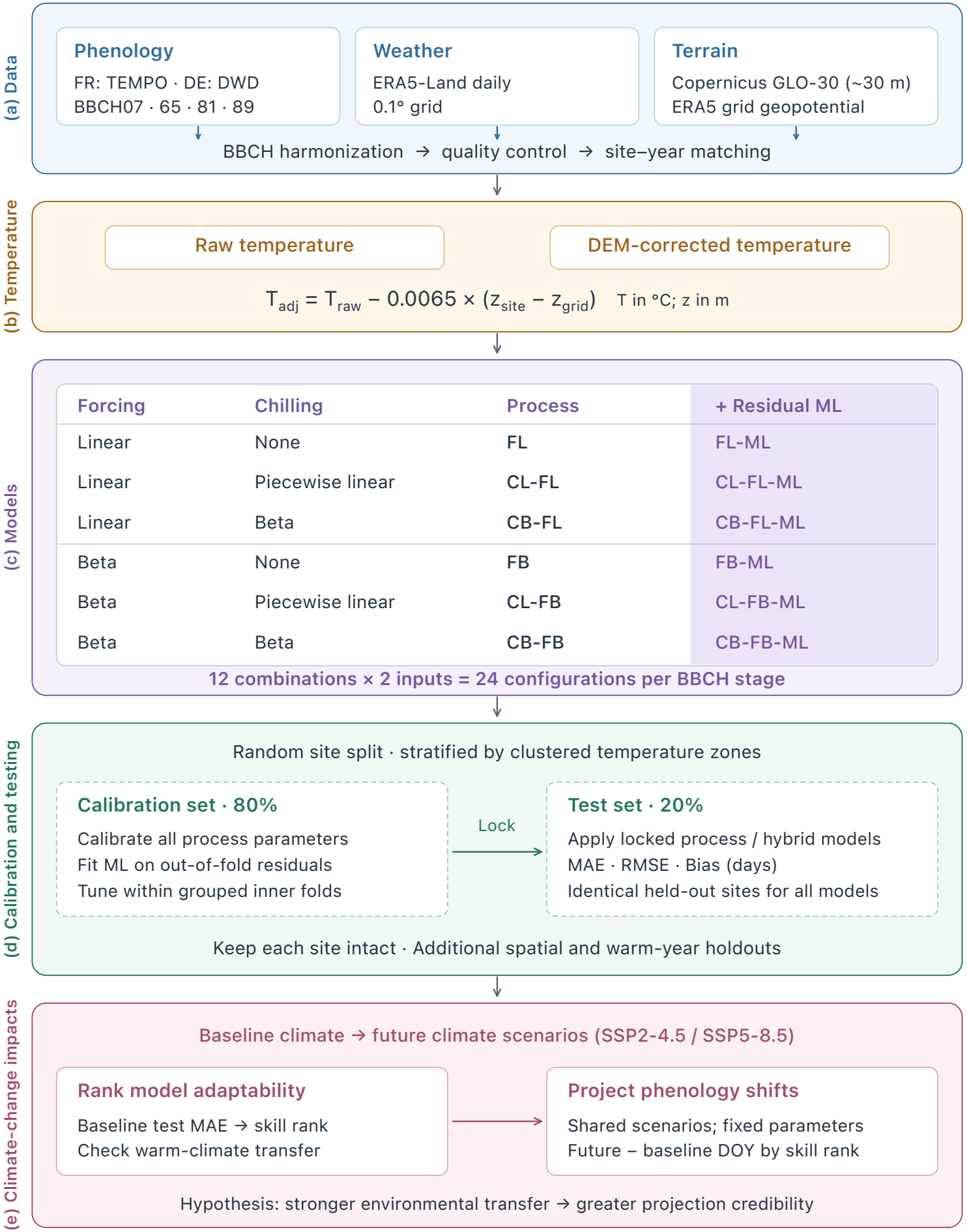
(a) Data
Phenology
FR: TEMPO · DE: DWD
BBCH07 · 65 · 81 · 89
Weather
ERA5-Land daily
0.1° grid
Terrain
Copernicus GLO-30 (~30 m)
ERA5 grid geopotential
BBCH harmonization → quality control → site–year matching
(b) Temperature
Raw temperature
DEM-corrected temperature
$T_{adj} = T_{raw} - 0.0065 \times (z_{site} - z_{grid})$ T in °C; z in m
(c) Models
Forcing | Chilling | Process | + Residual ML
Linear | None | FL | FL-ML
Linear | Piecewise linear | CL-FL | CL-FL-ML
Linear | Beta | CB-FL | CB-FL-ML
Beta | None | FB | FB-ML
Beta | Piecewise linear | CL-FB | CL-FB-ML
Beta | Beta | CB-FB | CB-FB-ML
12 combinations × 2 inputs = 24 configurations per BBCH stage
(d) Calibration and testing
Random site split · stratified by clustered temperature zones
Calibration set · 80%
Calibrate all process parameters
Fit ML on out-of-fold residuals
Tune within grouped inner folds
Lock
Test set · 20%
Apply locked process / hybrid models
MAE · RMSE · Bias (days)
Identical held-out sites for all models
Keep each site intact · Additional spatial and warm-year holdouts
(e) Climate-change impacts
Baseline climate → future climate scenarios (SSP2-4.5 / SSP5-8.5)
Rank model adaptability
Baseline test MAE → skill rank
Check warm-climate transfer
Project phenology shifts
Shared scenarios; fixed parameters
Future – baseline DOY by skill rank
Hypothesis: stronger environmental transfer → greater projection credibility